

## Project Design Phase-II Data Flow Diagram & User Stories

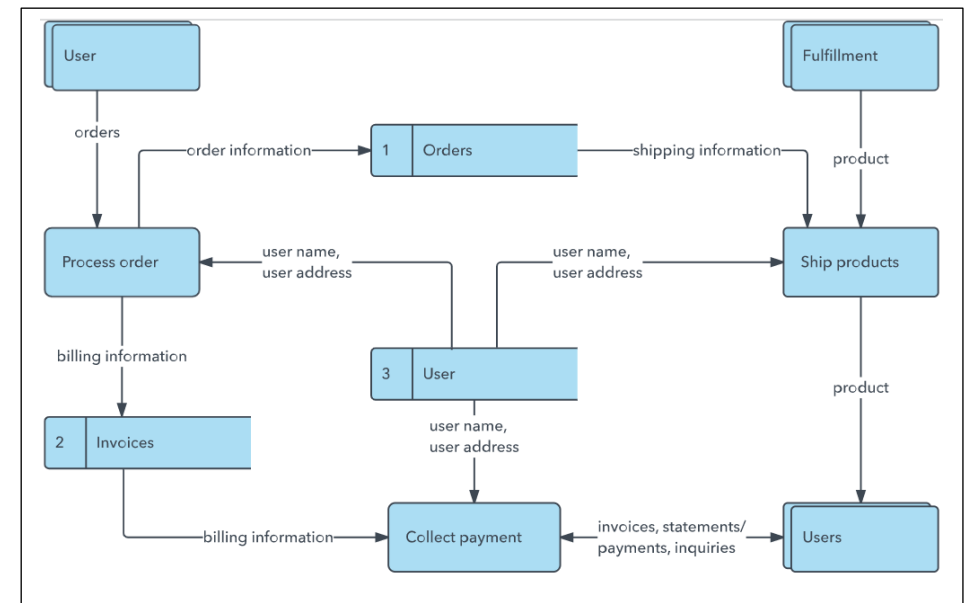
|               |                                                                                    |
|---------------|------------------------------------------------------------------------------------|
| Date          | 22 July 2026                                                                       |
| Team ID       | TEAM-HT-2026                                                                       |
| Project Name  | Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau |
| Maximum Marks | 4 Marks                                                                            |

### Data Flow Diagrams:

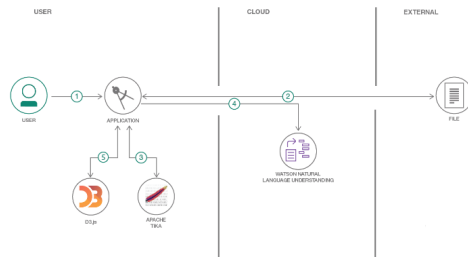
A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Example: [\(Simplified\)](#)

Example: DFD Level 0 (Industry Standard)



## Flow



1. User configures credentials for the Watson Natural Language Understanding service and starts the app.
2. User selects data file to process and load.
3. Apache Tika extracts text from the data file.
4. Extracted text is passed to Watson NLU for enrichment.
5. Enriched data is visualized in the UI using the D3.js library.

## User Stories

Use the below template to list all the user stories for the product.

| User Type          | Functional Requirement (Epic) | User Story Number | User Story / Task                                                                   | Acceptance criteria                                                            | Priority | Release  |
|--------------------|-------------------------------|-------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|----------|----------|
| Analyst / End User | Data Access & Exploration     | USN-1             | As an analyst, I can view a treemap of countries sized by number of heritage sites. | Treemap loads with correct counts per country from Country and Name_en fields. | High     | Sprint-1 |
|                    |                               | USN-2             | As an analyst, I can see a pie chart of sites 'In Danger' vs 'Not in Danger'.       | Pie chart accurately reflects proportion using the Danger field.               | High     | Sprint-1 |
|                    |                               | USN-3             | As an analyst, I can view a line chart of                                           | Line chart updates                                                             | High     | Sprint-1 |

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|------------------------------|-------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------|----------|
|                              |                               |                   | inscriptions per year, segmented by Region.                                                                    | correctly using Date_inscribed and Region fields.                                        |          |          |
|                              | Filtering & Interaction       | USN-4             | As a user, I can filter the dashboard by Region, Category and Year range.                                      | Selecting a filter updates all three visualizations simultaneously.                      | High     | Sprint-2 |
|                              |                               | USN-5             | As a user, I can hover over a country block to see the exact site count and site names.                        | Tooltip shows accurate country-level detail on hover.                                    | Medium   | Sprint-2 |
| Tourism Board / Policy Maker | Story Navigation              | USN-6             | As a policy maker, I can move through a guided story (Overview → At-Risk Sites → Regional Trends → Takeaways). | Story points navigate in order and each scene highlights the relevant chart.             | Medium   | Sprint-2 |
|                              | Data Quality                  | USN-7             | As a stakeholder, I trust that the dataset has been cleaned of duplicate or null entries before visualization. | No blank/duplicate Country, Region, or Danger values appear in any chart.                | High     | Sprint-1 |
| Administrator / Developer    | Publishing & Web Integration  | USN-8             | As an admin, I can publish the workbook to Tableau Server/Public and embed it in a web page.                   | Dashboard and story render correctly inside the Flask-hosted page.                       | High     | Sprint-3 |
|                              |                               | USN-9             | As an admin, I can monitor how much data is loaded and confirm filters/calculated fields perform without lag.  | Dashboard loads within an acceptable time with all filters and calculated fields active. | Medium   | Sprint-3 |

## Our Data Flow Diagram – Heritage Treasures Project

The Level-0 DFD below shows how data moves from the raw UNESCO dataset through cleaning and calculated fields, into Tableau visualizations, and finally to the published dashboard and story that end users and policy makers interact with.

